

day 19 | 251010 F

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def bisection(function, lower_guess, upper_guess,
tolerance=2**(-32)):
    midpoint = (lower_guess + upper_guess)/2
    while upper_guess - lower_guess > tolerance:
        if function(lower_guess)*function(midpoint)<0:
            upper_guess = midpoint
            midpoint = (lower_guess + upper_guess)/2
        elif function(midpoint)*function(upper_guess)<0:
            lower_guess = midpoint
            midpoint = (lower_guess + upper_guess)/2
        elif function(lower_guess)*function(midpoint)>0 and
function(midpoint)*function(upper_guess)>0:
            print('no unique root in that bracket')
            break
    return(midpoint)

def secant(f, guess, delta, tolerance = 2**(-32)):
    x0 = guess
    x1 = x0 + delta
    n = 0
    while abs(f(x1))>tolerance:
        x1 = x1 - (x1-x0)/(f(x1)-f(x0))*f(x1)
        n += 1
    return(x1, n)
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In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
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In [2]: def bisection(function, lower_guess, upper_guess, tolerance=2**(-32)):
midpoint = (lower_guess + upper_guess)/2
while upper_guess - lower_guess > tolerance:
    if function(lower_guess)*function(midpoint)<0:
        upper_guess = midpoint
        midpoint = (lower_guess + upper_guess)/2
    elif function(midpoint)*function(upper_guess)<0:
        lower_guess = midpoint
        midpoint = (lower_guess + upper_guess)/2
    elif function(lower_guess)*function(midpoint)>0 and function(midpoint)*function(upper_guess)>0:
        print('no unique root in that bracket')
        break
    return(midpoint)

def secant(f, guess, delta, tolerance = 2**(-32)):
    x0 = guess
    x1 = x0 + delta
    n = 0
    while abs(f(x1))>tolerance:
        x1 = x1 - (x1-x0)/(f(x1)-f(x0))*f(x1)
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    n += 1  
    return(x1, n)
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In []: